

6(a)(i)	<i>domain:</i> Eukarya ; <i>genus:</i> <u>Ailurus</u> and <i>species:</i> (<i>Ailurus</i> / <i>A.</i>) <u>fulgens</u> ;	2	ALLOW Eukaryota / eukaryote(s) / phonetic spelling CON for small 'a' for <i>Ailurus</i> or big 'F' for <i>fulgens</i>
6(a)(ii)	1 similar / shared, (described) morphological features / DNA sequences, to, other Carnivora / others in that order ; 2 shared, common ancestor / phylogeny / evolutionary relationships (with, other Carnivora / others in that order) ;	2	1 ALLOW share, physical / anatomical / biochemical, features / traits / characteristics with Carnivora
6(b)	<i>any three from:</i> <i>(yes / successful, because):</i> 1 (red panda species) has not, died out / gone extinct OR genetic variation preserved ; <i>(no / unsuccessful, because):</i> 2 habitat loss / wild population decline, has, continued / not stopped ; 3 few reintroductions have occurred / reintroduction only started in 2011 ; 4 AVP ;	3	IGNORE advice for future actions 1 ALLOW insurance population maintained 2 ALLOW habitats not, conserved / protected / restored 4 e.g. captive bred / zoo bred / released, animals may not survive in wild

Question	Answer	Marks	Guidance
6(c)	any five from: 1 increase <u>birth rate</u> / overcome reluctance to mate ; 2 IVF / in vitro fertilisation / fertilisation in test tube ; 3 hormone / FSH / LH, so produce many (mature), eggs / oocytes / follicles OR hormone / FSH/ LH, for superovulation / to increase ovulation ; 4 harvest, oocytes / eggs (from, ovary / oviduct), and sperm(atazoa) ; 5 embryo transfer / introduce embryo, to <u>uterus</u> ; 6 hormone / oestrogen / progesterone, prepares, uterus lining / endometrium ; 7 correct <i>ref. to</i> surrogate, mother / female ; 8 of, same / related, species ; 9 AVP ; 10 AVP ;	5	2 ALLOW petri dish for 'test tube' 4 ALLOW take gametes from both male and female 7 CON in wrong context e.g. AI / SCNT 8 ALLOW another red panda (female) 9, 10 e.g. artificial insemination / AI / semen placed in uterus embryo screening / select healthy embryos to implant zoos exchange, sperm / embryos, for genetic diversity OR frozen, ova / semen, from dead pandas, for genetic diversity OR sperm bank for, genetic diversity / store alleles idea